ADVANCED PROGRAMMING PROJECT REPORT

Ink Marker Production Line Simulator

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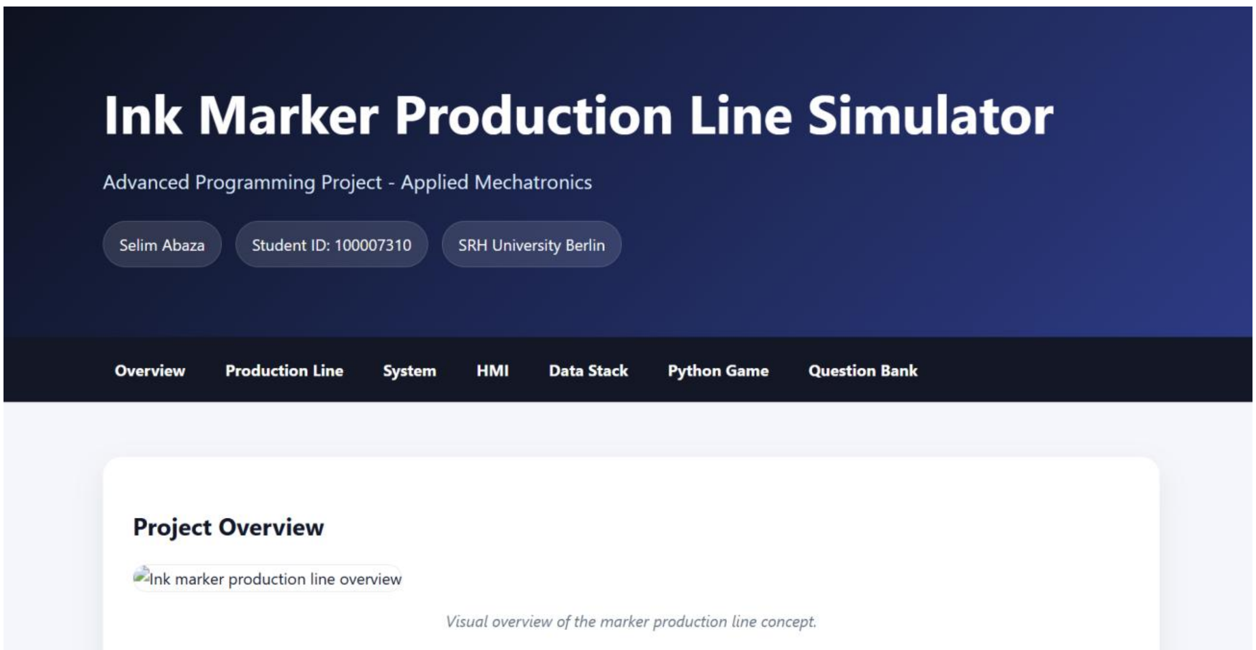

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1. INTRODUCTION

This project presents an Ink Marker Production Line Simulator developed as part of the Advanced Programming course. The system demonstrates how modern manufacturing processes can be modeled using object-oriented programming, industrial automation concepts, and software engineering principles.

The simulator represents a complete production process where ink markers move through multiple stations before reaching final quality control. The project combines backend production logic, a graphical Human Machine Interface (HMI), telemetry generation, data monitoring concepts, and web publication through GitHub Pages.

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2. PROJECT OBJECTIVES

The primary objectives of the project were:

• Develop a modular production line simulator using Python.
• Apply object-oriented programming principles.
• Create a graphical HMI for operator interaction.
• Simulate industrial manufacturing processes.
• Demonstrate defect detection and quality control.
• Integrate telemetry and monitoring concepts.
• Use Git and GitHub for version control.
• Publish project documentation through GitHub Pages.

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3. PRODUCTION LINE DESCRIPTION

The Ink Marker Production Line consists of five manufacturing stages:

1. Marker Body Preparation

2. Ink Filling Station

3. Tip Installation
4. Cap Assembly
5. Quality Control and Packaging

Each marker progresses through the production sequence while the system tracks successful products and defective products. Defects may include underfilled reservoirs, damaged bodies, missing tips, or faulty caps.

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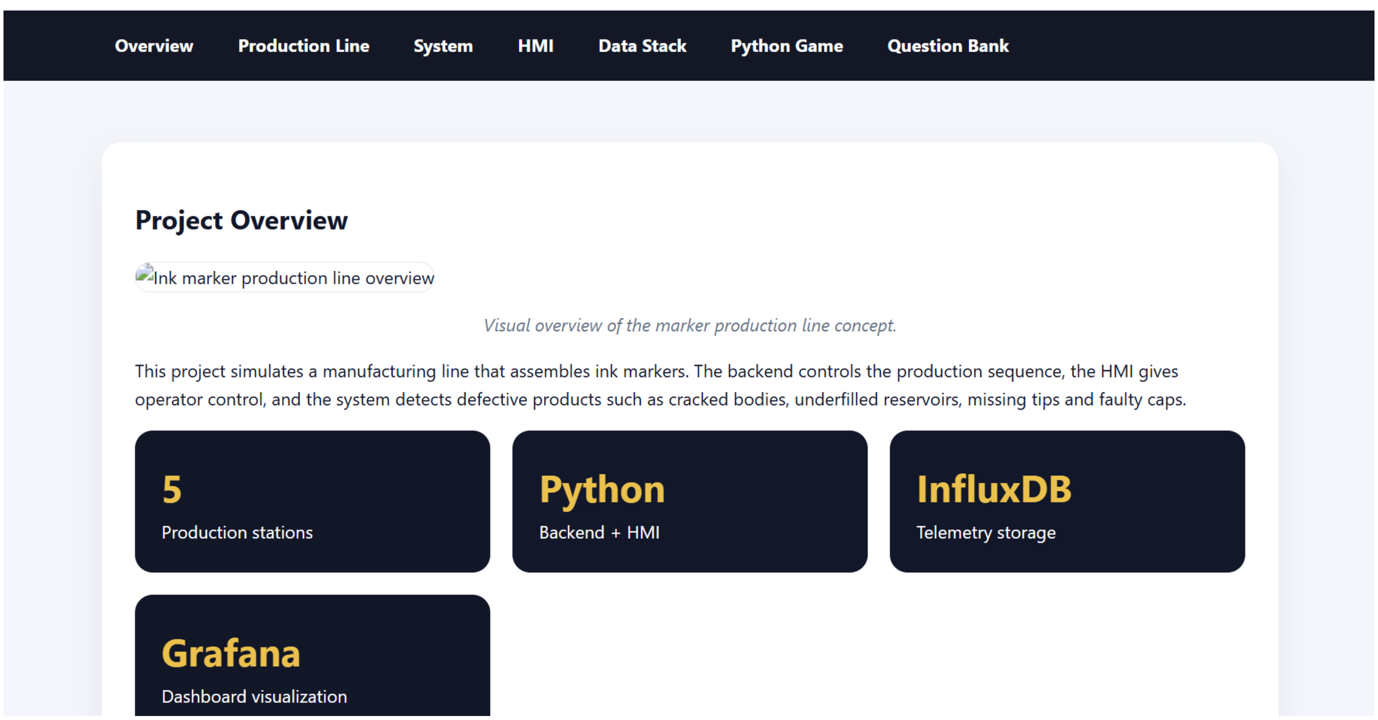
4. SYSTEM ARCHITECTURE

The software architecture follows a modular structure consisting of:

• Frontend HMI
• Production Controller
• Station Classes
• Product Models
• Telemetry Generator
• Monitoring Layer

This separation improves maintainability, readability, and scalability.

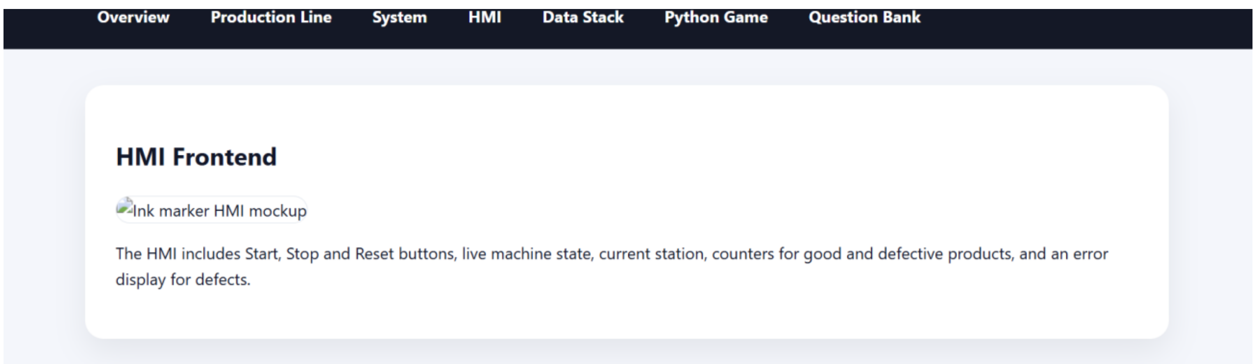


5. HUMAN MACHINE INTERFACE (HMI)

The Human Machine Interface provides:

- Start button
- Stop button
- Reset button
- Machine status display
- Current production stage display
- Good product counter
- Defective product counter
- Error and fault messages

The HMI allows an operator to monitor production performance in real time.



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6. SOFTWARE STRUCTURE

main.py
Application entry point responsible for launching the simulator.

production_line.py
Contains the main production sequence logic and coordination between stations.

stations.py
Defines manufacturing stations and processing behavior.

models.py
Contains data structures representing markers and production objects.

database.py
Handles telemetry and data storage interactions.

producer.py
Generates production data used for monitoring and visualization.

hmi.py
Implements the graphical Human Machine Interface.

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7. GIT AND GITHUB USAGE

Git was used as the version control system throughout development. It enabled tracking of project changes, maintaining file history, and managing updates efficiently.

GitHub was used to:

- Host the repository
- Store project files
- Publish documentation
- Provide project sharing and submission links

Repository Link:

https://github.com/Selim027/100007310_Abaza_AP

Website Link:

https://selim027.github.io/100007310_Abaza_AP/

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8. DOCKER, INFLUXDB AND GRAFANA

Docker was included to support containerized deployment of monitoring services.

InfluxDB serves as a time-series database used to store production telemetry data.

Grafana provides dashboards and visualizations for monitoring production performance, throughput, and defect statistics.

These technologies demonstrate industrial monitoring concepts commonly used in manufacturing environments.

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9. AI TOOLS USAGE

AI tools such as ChatGPT were used during development for:

• Project planning
• Debugging support
• Code explanations
• Documentation assistance
• Website content refinement

All generated material was reviewed, modified, tested, and integrated manually before submission.

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10. TESTING AND RESULTS

Testing activities included:

• Verification of production flow
• HMI functionality testing
• Defect detection validation
• Website deployment verification
• Repository accessibility verification

Results demonstrated successful operation of the simulated production process and correct interaction between system components.

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11. CONCLUSION

The Ink Marker Production Line Simulator successfully demonstrates industrial automation concepts using modern software engineering techniques. The project combines object-oriented programming, user interface development, monitoring technologies, version control, and web publication into a single integrated solution.

The completed project satisfies the objectives of the Advanced Programming assignment while providing a realistic simulation of an industrial manufacturing process.

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12. REFERENCES

Python Documentation:
https://www.python.org

Git Documentation:
https://git-scm.com

GitHub Documentation:
https://docs.github.com

Docker Documentation:
https://docs.docker.com

InfluxDB Documentation:
https://docs.influxdata.com

Grafana Documentation:
https://grafana.com/docs